Package ‘limai’

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Title Linear modeling and AI decision making

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Author Xu Liu [aut,cre]

Maintainer Xu Liu <liu.xu@sufe.edu.cn>

Description Datasets used in the book ``Linear modeling and AI decision making"

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Depends R (>= 3.5.0)

Imports Matrix, glmnet

LazyData true

Contents

`apprentice` *Apprentice migration to Edinburgh*

Description

The number of apprentices migrating to Edinburgh

Usage

```
data(apprentice)
```

Format

A data frame with 33 observations on the following 5 variables.

`Dist` the distance from Edinburgh (unit unknown, presumably miles); a numeric vector

`Apps` the number of apprentices moving to Edinburgh from the given county (given in row labels); a numeric vector

`Pop` the population (in thousands) of the given county; a numeric vector

`Urban` the degree of urbanization as measured by the percentage of the population living in urban settlements; a numeric vector

`Locn` the location of the county relative to Edinburgh; a factor with levels `North`, `South` and `West`

Details

The data record the number of apprentices moving to Edinburgh between 1775 and 1799 from other Scottish counties.

Source

Andrew Lovett and Robin Flowerdew (1989) Analysis of count data using Poisson regression. *Professional Geographer*, **41**(2), 190–198.

R package GLMsData: https://cran.r-project.org/web/packages/GLMsData/index.html

Examples

```
data(apprentice)
summary(apprentice)
```

`boston` *Boston housing data*

Description

The dataset contains information on housing values in suburbs of Boston, including various attributes such as crime rate, property tax, and average number of rooms.

Usage

```
data(boston)
```

Arguments

`crim`	Per capita crime rate by town.
`zn`	Proportion of residential land zoned for large lots (over 25,000 square feet).
`indus`	Proportion of non-retail business acres per town.
`chas`	Charles River dummy variable (1 if tract bounds river; 0 otherwise).
`nox`	Nitrogen oxide concentration (parts per 10 million).
`rm`	Average number of rooms per dwelling.
`age`	Proportion of owner-occupied units built before 1940.
`dis`	Weighted distance to employment centers in Boston.
`rad`	Index of accessibility to radial highways.
`tax`	Full-value property tax rate per $10,000.
`ptratio`	Pupil-teacher ratio by town.
`b`	Proportion of residents of African American descent.
`lstat`	Percentage of lower status population.
`medv`	Median value of owner-occupied homes in $1000s.

Details

The dataset is derived from the Boston Housing dataset, originally from the UCI Machine Learning Repository. It contains data collected from 506 census tracts in Boston, providing a snapshot of various housing-related features, which can be used for regression and classification tasks in machine learning.

Source

https://lib.stat.cmu.edu/datasets/boston or R package MASS

References

Harrison, D., & Rubinfeld, D. L. (1978). Hedonic housing prices and the demand for clean air. *Journal of Environmental Economics and Management*, **5**(1), 81-102.

Examples

```
data(boston)
model = lm(medv ~ ., data = boston)
summary(model)
```

`breastcancer` *Breast cancer data*

Description

The dataset contains gene expression and gene copy number information from 89 subjects.

Usage

```
data(breastcancer)
```

Arguments

`dna`	Copy number variation (CNV) data representing genomic DNA amplification or deletion events in tumor samples.
`rna`	Gene expression profiles measured via RNA transcript levels (e.g., microarray or RNA-seq data).
`chrom`	Chromosome numbers (1-22, X, Y) corresponding to the genomic location of the measured genes.
`nuc`	Nucleotide positions (start/end coordinates) of genes or probes on the chromosome (e.g., hg18/hg19 reference).
`gene`	Unique gene identifiers (e.g., Entrez Gene IDs or probe IDs) linked to genomic features.
`genenames`	Official gene symbols or names (e.g., BRCA1, ERBB2) standardized by HUGO Gene Nomenclature Committee (HGNC).
`genechr`	Chromosomal mapping information for each gene (e.g., "chr17" for TP53).
`genedesc`	Brief functional descriptions of genes (e.g., "tumor protein p53" or "estrogen receptor 1").
`genepos`	Genomic coordinates of genes (e.g., cytoband or base-pair positions like "17q21.31").

Details

The dataset is derived from molecular bioinformatics data obtained from breast cancer tissue samples treated according to the standard of care between 1989 and 1997. It primarily consists of gene expression profiles and copy number variation data across 22 chromosomal pairs in tumor tissue samples from 89 breast cancer patients. For a detailed explanation of this dataset, please refer to Chin et al. (2006).

Source

http://icbp.lbl.gov/breastcancer/ or R package PMA

References

Chin, K., DeVries, S., et al. (2006). Genomic and transcriptional aberrations linked to breast cancer pathophysiologies. *Cancer cell*, **10**(6), 529-541.

Examples

```
library(glmnet)
data(breastcancer)
dna = breastcancer$dna[breastcancer$chrom==21,]
rna = breastcancer$rna[which(breastcancer$genechr==21),]
y = dna[1,]
x = t(rna)
set.seed(100)
fit_ridge = cv.glmnet(x,y,alpha = 0)
coef(fit_ridge, s = "lambda.min")
fit_lasso = cv.glmnet(x,y,alpha = 1)
coef(fit_lasso, s = "lambda.min")
```

`cars` *Car Speed and Braking Distance Dataset*

Description

A classic dataset recording the relationship between car speed and corresponding braking distance for automobiles in the 1920s. It contains 50 samples with two variables: one continuous feature (speed) and one continuous target variable (braking distance). This dataset is widely used for exploring linear relationships between two continuous variables, particularly in introductory linear regression analysis and statistical modeling tutorials.

Usage

```
data(cars)
```

Arguments

A data frame object containing 50 observations with the following 2 variables:

Variable Name	**Variable Description**
speed	Car speed (measured in miles per hour; feature variable)
dist	Braking distance (measured in feet; target variable)

Details

The dataset captures real-world measurements of automobile performance, focusing on how braking distance increases with speed. Each sample represents a unique observation of a car's speed and the distance required to come to a complete stop. It is particularly useful for demonstrating basic linear regression concepts (e.g., fitting a linear model to predict distance from speed) and visualizing bivariate relationships (e.g., scatter plots with regression lines). The simplicity of the dataset (two variables, no missing values) makes it ideal for teaching introductory statistics and machine learning.

Source

Built-in dataset in R's base datasets package (included with base R)

References

McNeil, D. R. (1977). *Interactive Data Analysis*. Wiley.
Venables, W. N., & Ripley, B. D. (2002). *Modern Applied Statistics with S*. Springer.

Examples

```
data(cars)

speed_dist_model <- lm(dist ~ speed, data = cars)
summary(speed_dist_model)

plot(cars$speed, cars$dist,
     xlab = "Speed (mph)",
     ylab = "Braking Distance (ft)",
     main = "Speed vs. Braking Distance")
abline(speed_dist_model, col = "red", lwd = 2)
```

CHARLS — *China Health and Retirement Longitudinal Study (CHARLS) data of Hebei, Shandong, and Fujian provinces.*

Description

The dataset contains the CHARLS data collected in Hebei, Shandong, and Fujian provinces.

Usage

```
data(CHARLS)
```

Format

A list object containing the following 3 variables:

Name	Type	Description
`hebei`	matrix	The CHARLS data of Hebei province. A matrix with 257 rows and 50 columns including y and 49 covariates v1,...,v49.
`shandong`	matrix	The CHARLS data of Shandong province. A matrix with 413 rows and 50 columns including y and 49 covariates v1,...,v49.
`fujian`	matrix	The CHARLS data of Fujian province. A matrix with 167 rows and 50 columns including y and 49 covariates v1,...,v49.

Details

Each matrix containing `y`, `v1`-`v49` variables:

Name	Description
`y`	Annual support income of elderly people.
`v1`	Gender: `1` = `male`, `0` = `female`.
`v2`	Age.
`v3`	Education level: `1` = `primary`, `2` = `junior high`, `3` = `high school`, `4` = `other`.
`v4`	Marital status: `1` = `married`, `0` = `unmarried`.

`v5`	Live alone: `1 = yes, 0 = no`.
`v6`	Live with a spouse: `1 = yes, 0 = no`.
`v7`	Live with children: `1 = yes, 0 = no`.
`v8`	Live with other members (e.g., parents): `1 = yes, 0 = no`.
`v9`	Health status: `1 = disability/chronic illness, 0 = healthy`.
`v10`	Pension income.
`v11`	Whether to receive a pension: `1 = yes, 0 = no`.
`v12`	Number of surviving children: `0 = none, 1 = one, 2 = two or more`.
`v13`	Wage income per household.
`v14`	Net operating income per household.
`v15`	Net transfer income per household.
`v16`	Number of children with a college degree or above.
`v17`	Number of children earning over 10,000 CNY annually.
`v18`	Emotional comfort: `1 = contact children \geq every half month, 0 = otherwise`.
`v19`	Number of household members.
`v20`	Number of deceased biological children.
`v21`	Number of surviving adopted children.
`v22`	Number of surviving sons.
`v23`	Financial support for parents.
`v24`	Financial support for other relatives.
`v25`	Net financial support received from other relatives.
`v26`	Number of types of disability.
`v27`	Chronic illness: `0 = no, 1 = yes, 2 = other`.
`v28`	Whether to receive a retirement pension: `1 = yes, 0 = no`.
`v29`	Retirement pension income.
`v30`	New rural pension income.
`v31`	All other pension income.
`v32`	Pension income of elderly households.
`v33`	Total financial assets of elderly and spouses.
`v34`	Wage income of main household members.
`v35`	Government subsidies for individual families.
`v36`	Government subsidies for main household members.
`v37`	Wage income of other family members.
`v38`	Government subsidies for other family members.
`v39`	Total government subsidies for families.
`v40`	Government transfer income for households.
`v41`	Net household income excluding private transfers.
`v42`	Net household income.
`v43`	Net household income per capita.
`v44`	Other net private transfer income of elderly.
`v45`	Family shared income received by elderly.
`v46`	Whether to complete junior high school education: `1 = yes, 0 = no`.
`v47`	Whether to complete high school education: `1 = yes, 0 = no`.
`v48`	Annual net income from other sources.
`v49`	Financial support provided for children.

Source

The CHARLS data from https://charls.charlsdata.com/pages/Data/2015-charls-wave4/zh-cn.html

References

Ren, P., Liu, X., Zhang, X., Zhan, P., & Qiu, T. (2024). Integrative analysis of high-dimensional quantile regression with contrasted penalization. *Journal of Applied Statistics*, 1-17.

Examples

```
library(glmnet)
data(CHARLS)
data_hebei = CHARLS$hebei
y = data_hebei$y
x = data_hebei[,-1]
x = matrix(unlist(x),nrow = nrow(x))
fit_lasso = cv.glmnet(x,y,alpha = 1)
coef(fit_lasso,s = "lambda.min")
```

`Concrete` *Concrete Slump Test Dataset*

Description

A classic dataset in materials science and civil engineering for studying the relationship between concrete mix proportions and its workability/mechanical properties. It contains 103 concrete samples, recording 7 raw material proportions and 3 key performance indicators. Widely used in multi-target regression modeling and material performance prediction research, as the three response variables (SLUMP, FLOW, and Compressive.Strength) need to be considered simultaneously in engineering practice and have inherent physical connections and trade-offs.

Usage

```
data(Concrete)
```

Arguments

A data frame containing 103 observations with the following 10 variables:

Name	**Description**
Cement	Cement content (kg/m^3; main cementitious material in concrete)
Slag	Slag content (kg/m^3; industrial by-product used as admixture)
Fly.ash	Fly ash content (kg/m^3; improves workability and durability)
Water	Water content (kg/m^3; affects workability and strength)
SP	Superplasticizer content (kg/m^3; enhances concrete fluidity)
Coarse.Aggr.	Coarse aggregate content (kg/m^3; primary skeleton material in concrete)
Fine.Aggr.	Fine aggregate content (kg/m^3; fills voids between coarse aggregates)
SLUMP	Slump (cm; indicator of fresh concrete fluidity)
FLOW	Flow (cm; final spread diameter of fresh concrete)
Compressive.Strength	28-day compressive strength (MPa; strength of hardened concrete)

Details

The dataset focuses on quantifying how the proportions of raw materials (cement, slag, fly.ash, water, superplasticizer, aggregates) influence concrete's workability (measured by SLUMP and FLOW) and mechanical strength (Compressive.Strength). These three response variables are interconnected in engineering: fluidity (SLUMP, FLOW) affects construction efficiency, while compressive.strength determines structural performance, and there are often trade-offs between workability and strength. This makes the dataset ideal for multi-response regression analysis, where multiple target variables are modeled simultaneously to capture their joint variation and inherent physical relationships.

Source

UCI Machine Learning Repository: https://archive.ics.uci.edu/ml/datasets/Concrete+Slump+Test

References

Yeh, I.-C. (2007). Modeling of strength of high-performance concrete using artificial neural networks. *Cement and Concrete Research*, 37(12), 1797-1808.

Examples

```
data(Concrete)

## Fit a simple linear regression model for SLUMP using Water content
fit_slump <- lm(SLUMP ~ Water, data = Concrete)
summary(fit_slump)

## Visualization: Slump vs. Water Content
plot(Concrete$Water, Concrete$SLUMP,
    xlab = "Water Content (kg/m^3)",
     ylab = "Slump (cm)",
     main = "Slump vs. Water Content")

abline(fit_slump, lwd = 2)
```

`emeraldaug` *August monthly rainfall in Emerald*

Description

The total monthly rainfall in Emerald, Australia, and the average monthly SOI

Usage

```
data(emeraldaug)
```

Format

A data frame with 114 observations on the following 3 variables.

`Year` the year; a numeric vector

`Rain` the total monthly rainfall in August of the given year; a numeric vector

`SOI` the monthly average southern oscillation index (SOI); a numeric vector

`Phase` the SOI phase (see Stone and Auliciems, 1992); a factor with these values: `1` (consistently negative), `2` (consistently positive), `3` (rapidly falling), `4` (rapidly rising), or `5` (consistently near zero)

Details

The data give the total monthly rainfall and monthly in Emerald, Queensland, Australia, from 1889 to 2002, and the average SOI for the corresponding month.

Source

Data obtained from the Australian Bureau of Meteorology (`http://www.bom.gov.au`) and IRI/LDEO Climate Data Library (http://www.longpaddock.qld.gov.au/seasonalclimateoutlook/southernoscillationindex/soidatafiles/ on 21 December 2010, then compiled.

R package GLMsData: https://cran.r-project.org/web/packages/GLMsData/index.html

References

R. C. Stone and A. Auliciems (1992) SOI phase relationships with rainfall in eastern Australia, *International Journal of Climatology*, **12**, 625–636.

Examples

```
data(emeraldaug)
plot(emeraldaug)
```

`Energy` *Energy Efficiency Dataset*

Description

A classic multi-response regression dataset for building energy consumption prediction, sourced from the UCI Machine Learning Repository. It contains 768 samples with 10 attributes: 8 independent variables describing building physical parameters (e.g., compactness, surface area) and 2 continuous target variables (Heating Load and Cooling Load) for simultaneous prediction. This dataset is widely used in validating multi-response linear regression models in building energy efficiency research.

Usage

```
data(Energy)
```

Arguments

A data frame object containing 768 observations with the following 10 variables:

Name	Type	Description
X1	continuous	Relative Compactness (core indicator of building compactness)
X2	continuous	Surface Area (total building surface area, unit: m^2)
X3	continuous	Wall Area (total building wall area, unit: m^2)
X4	continuous	Roof Area (total building roof area, unit: m^2)
X5	continuous	Overall Height (total building height, unit: m)
X6	integer	Orientation (building orientation, 1–4 representing directions)
X7	continuous	Glazing Area (proportion of glass area to total wall area)
X8	integer	Glazing Area Distribution (glass layout pattern, 1–4)
Y1	continuous	Heating Load (annual heating energy, kWh/m^2; target)
Y2	continuous	Cooling Load (annual cooling energy, kWh/m^2; target)

Details

The dataset focuses on quantifying relationships between building design features and energy consumption, providing a foundation for energy-efficient building design and HVAC system optimization. Each sample represents a unique building configuration, enabling simultaneous prediction of heating and cooling demands—two key metrics for evaluating building energy performance. It supports academic research (e.g., multi-response regression, feature importance analysis) and industrial applications (e.g., pre-design energy forecasting, energy audit).

Source

UCI Machine Learning Repository: https://archive.ics.uci.edu/ml/datasets/Energy+Efficiency

References

Tsanas, A., & Xifara, A. (2012). Accurate quantitative estimation of energy performance of residential buildings using statistical machine learning tools. *Energy and Buildings*, 49, 560-567.

Examples

```
data(Energy)

## Fit two separate linear regression models
fit_Y1 <- lm(Y1 ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8, data = Energy)
fit_Y2 <- lm(Y2 ~ X1 + X2 + X3 + X4 + X5 + X6 + X7 + X8, data = Energy)

summary(fit_Y1)
summary(fit_Y2)

## Simple visualization: Cooling Load vs. Glazing Area
plot(Energy$X7, Energy$Y2,
     xlab = "Glazing Area",
     ylab = "Cooling Load",
     main = "Cooling Load vs. Glazing Area")

abline(lm(Y2 ~ X7, data = Energy), lwd = 2)
```

`fluoro` *The time of fluoroscopy and total radiation*

Description

The data give the total procedure time during CT fluoroscopic scanning, and the radiation dose received.

Usage

```
data(fluoro)
```

Format

A data frame with 19 observations on the following 2 variables.

`Time` the total procedure time (in minutes); a numeric vector

`Dose` the total radiation dose received (in rads); a numeric vector

Details

The data are given in the Table as the natural log of `Time` and the natural log of `Dose`. Here the data have been transformed back to the original scale. The source claims the purpose of the data collection was "to assess whether radiation dose could be estimated by simply measuring the total CT fluoroscopic procedure time". The procedure was performed in the abdomen.

Source

Kelly H. Zou, Kemal Tuncali, and Stuart G. Silverman (2003) Correlation and simple linear regression. *Radiology*, **227**, 617–628.

R package GLMsData: https://cran.r-project.org/web/packages/GLMsData/index.html

References

The data were originally used, but not given, in S. G. Silverman, K. Tuncali, D. F. Adams, R. D. Nawfel, K. H. Zou, and P. F. Judy (1999) CT fluoroscopy-guided abdominal interventions: techniques, results, and radiation exposure. *Radiology*, **212**, 673–681.

Examples

```
data(fluoro)
plot(fluoro)
```

game *Online Gaming Behavior Dataset*

Description

This dataset captures comprehensive metrics and demographics related to player behavior in online gaming environments. It includes variables such as player demographics, game-specific details, engagement metrics, and a target variable reflecting player retention.

Usage

```
data(game)
```

Arguments

A data frame object containing 200 player entries with the following 13 variables:

Name	Type	Description
ID	integer	Unique identifier for each player
age	integer	Age of the player
gender	character	Gender of the player
location	character	Geographic location of the player
genre	character	Genre of the game the player is engaged in
time	numeric	Average hours spent playing per session
inbuy	integer	Indicates whether the player makes in-game purchases (0 = No, 1 = Yes)
difficulty	character	Difficulty level of the game
session	integer	Number of gaming sessions per week
sesslong	integer	Average duration of each gaming session in minutes
level	integer	Current level of the player in the game
achievement	integer	Number of achievements unlocked by the player
engagement	character	Categorized engagement level reflecting player retention ('High', 'Medium', 'Low')

Details

The data provides information on various aspects of online gaming behavior for 200 players, including player identifiers, demographic details, gaming - related metrics.

Source

The dataset from Kaggle website https://www.kaggle.com/datasets/rabieelkharoua/predict-online-gaming-behavior-dataset/data

Examples

```
data(game)

# Check the relationship between in - game purchases and session length
cor(game$inbuy, game$sesslong, use = "complete.obs")

# Count the number of players in each game genre
genre_counts <- table(game$genre)
```

```
barplot(genre_counts,
        main = "Number of Players in Each Game Genre",
        xlab = "Game Genre",
        ylab = "Number of Players")
```

gpa1 *GPA and Student Characteristics Dataset*

Description

A classic dataset in educational economics and econometric teaching, containing 141 observations on 29 student characteristics collected at Michigan State University (MSU) in 1994.

Usage

```
data(gpa1)
```

Format

A data frame with 141 observations on 29 variables:

age age in years

soph equals 1 if sophomore

junior equals 1 if junior

senior equals 1 if senior

senior5 equals 1 if fifth-year senior

male equals 1 if male

campus equals 1 if living on campus

business equals 1 if business major

engineer equals 1 if engineering major

colGPA college GPA at MSU

hsGPA high school GPA

ACT ACT achievement score

job19 equals 1 if job hours <= 19

job20 equals 1 if job hours >= 20

drive equals 1 if drives to campus

bike equals 1 if bicycles to campus

walk equals 1 if walks to campus

voluntr equals 1 if does volunteer work

PC equals 1 if personal computer at school

greek equals 1 if member of a Greek organization

car equals 1 if owns a car

siblings equals 1 if has siblings

bgfriend equals 1 if has boyfriend or girlfriend

clubs equals 1 if belongs to an MSU club

skipped average number of lectures missed per week

alcohol average number of days per week drinks alcohol

gradMI equals 1 if Michigan high school graduate

fathcoll equals 1 if father is a college graduate

mothcoll equals 1 if mother is a college graduate

Details

This dataset is commonly used to demonstrate linear regression modeling, hypothesis testing, and variable selection. It examines how academic background (high school GPA, ACT), demographics (gender, class standing), lifestyle choices (work hours, commuting, alcohol consumption), and family education relate to college GPA.

Source

Originally from Wooldridge (2019), *Introductory Econometrics*. Also available in the **Wooldridge** R package.

References

Wooldridge, J. M. (2019). *Introductory Econometrics: A Modern Approach* (7th ed.). Cengage.

Examples

```
data(gpa1)

gpa_model <- lm(colGPA ~ hsGPA + ACT, data = gpa1)
summary(gpa_model)

plot(gpa1$hsGPA, gpa1$colGPA,
     xlab = "High School GPA",
     ylab = "College GPA",
     main = "College GPA vs. High School GPA")
```

GTEx_apoe — *GTEx data associated with APOE gene*

Description

The dataset contains APOE gene and other related genes from 111 subjects.

Usage

```
data(GTEx_apoe)
```

Details

This dataset contains gene expression profiles and genomic data derived from the Genotype-Tissue Expression (GTEx) project. It is a list with 111 rows and 119 columns, and the last column is the APOE gene, the other 118 columns is the related genes.

Source

Genotype-Tissue Expression (GTEx) project, available at: https://www.gtexportal.org/home/

Examples

```
library(glmnet)
data(GTEx_apoe)
y = GTEx_apoe$APOE
x = GTEx_apoe[,-1]
x = matrix(unlist(x), nrow = nrow(x))
fit_lasso = cv.glmnet(x,y,alpha = 1)
coef(fit_lasso,s = "lambda.min")
```

---

happiness *World Happiness Report Dataset (2016)*

Description

This dataset provides country-level measures of happiness from the World Happiness Report 2016. It includes overall happiness scores, rankings, and contributing factors such as GDP, family support, life expectancy, freedom, generosity, and corruption perception. The dataset enables quantitative analysis of global well-being and policy-related research.

Usage

```
data(happiness)
```

Arguments

A data frame object containing country-level records with the following 13 variables:

Variable Name	**Variable Description**
Country	Country name
Region	Region where the country is located
Rank	Ranking based on overall happiness score
Score	Happiness score in 2016 (0-10)
Economy (GDP per Capita)	Contribution of GDP per capita to happiness score
Family	Contribution of social support to happiness score
Health (Life Expectancy)	Contribution of life expectancy to happiness score
Freedom	Contribution of personal freedom to happiness score
Generosity	Contribution of generosity to happiness score
Trust (Government Corruption)	Contribution of distrust in government corruption to happiness score
Dystopia Residual	Residual compared to the "dystopia" benchmark

Details

The World Happiness Report dataset is based on Gallup World Poll survey results, evaluating happiness scores across countries. The dataset allows exploration of how economic, social, and political factors contribute to happiness.

Source

Data from Kaggle: https://www.kaggle.com/datasets/unsdsn/world-happiness

Examples

```
data(happiness)

hist(happiness$happ_score,
     main = "Distribution of Happiness Scores (2016)",
     xlab = "Happiness Score",
     ylab = "Frequency")
```

`hcrabs` *Males attached to female horseshoe crabs*

Description

The number of male crabs attached to female horseshoe crabs

Usage

```
data(hcrabs)
```

Format

A data frame with 173 observations on the following 5 variables.

`Col` the color of the female; a factor with levels `LM` (light medium), `M` (medium), `DM` (dark medium) or `D` (dark)

`Spine` the spine condition; a factor with levels `BothOK`, `OneOK` or `NoneOK`

`Width` the carapace width of the female crab in cm; a numeric vector

`Wt` the weight of the female crab in grams; a numeric vector

`Sat` the number of male crabs attached to the female ('satellites'); a numeric vector

Details

The data come from an observational study of nesting horseshoe crabs: "The study was conducted at two beaches on the Delaware shore, Breakwater Harbor at Cape Henlopen Park in Lewes and Fowler's Beach, 32 km north on the same shoreline (Sussex County, Delaware, USA). In 1991 observations were made from 7 to 17 June, in1992 from 28 May to 3 June and from 11 to 14 June, and in 1993 from 18 May to 11 June. At these sites the crabs were most active on the higher of the two daily high tides (which at this time of year are at night between 1700 and 0200 h EST)" (Brockmann, 1996; p. 4).

Source

H. J. Brockmann (1996) Satellite male groups in horseshoe crabs, *Limulus polyphemus*. *Ethology*, **102**(1), 1–21.

R package GLMsData: https://cran.r-project.org/web/packages/GLMsData/index.html

Examples

```
data(hcrabs)
plot(Sat ~ Wt, data=hcrabs)
```

`insurance` *Medical Insurance Expenditure Dataset*

Description

This dataset contains annual medical charges for insured individuals in the United States, along with demographic and lifestyle-related predictors. It is commonly used for building predictive models (e.g., regression) to estimate future medical costs, which supports actuarial pricing and risk stratification.

Usage

```
data(insurance)
```

Arguments

A data frame with 1,338 observations of insured individuals (aged 18–64), with the following 7 variables:

Name	**Type**	**Description**
`age`	integer	Age of the primary beneficiary (18–64 years)
`sex`	character	Gender of the insured (female/male)
`bmi`	numeric	Body mass index (kg/m^2)
`children`	integer	Number of children/dependents covered by insurance
`smoker`	character	Smoking status (yes/no)
`region`	character	Residential region (northeast/southeast/southwest/northwest)
`charges`	numeric	Annual medical expenditure (in USD)

Details

The dataset was simulated based on U.S. Census Bureau data and introduced in Chapter 6 of *Machine Learning with R*. It is widely used for regression modeling to estimate healthcare costs and to demonstrate how demographic and lifestyle variables influence expenditures.

Source

Original dataset from GitHub: https://github.com/stedy/Machine-Learning-with-R-datasets

Examples

```
data(insurance)

# Display first rows
head(insurance)

# Fit a linear regression model: charges ~ age + bmi + children + smoker + region
model <- lm(charges ~ age + bmi + children + smoker + region,
```

```
            data = insurance)

summary(model)
```

`lungcapacity` *Children's Lung Capacity Dataset*

Description

This dataset contains measurements of lung capacity in children and adolescents aged 3 to 19 years. It includes physiological measurements, demographic information, and health-related factors that may influence respiratory development.

Usage

```
data(lungcapacity)
```

Arguments

A data frame with 725 observations and the following 6 variables:

Variable	**Type**	**Description**
`LungCap`	numeric	Lung capacity measurement (liters)
`Age`	integer	Age of child in years (3-19)
`Height`	numeric	Height in inches (45.3-81.8)
`Smoke`	character	Smoking habit (no/yes)
`Gender`	character	Gender (male/female)
`Caesarean`	character	Born by Caesarean section (no/yes)

Source

The dataset from Kaggle website https://www.kaggle.com/datasets/jacopoferretti/lung-capacity-of-kids

Examples

```
data(lungcapacity)

# Basic summary statistics
summary(lungcapacity)

# Compare lung capacity between smokers and non-smokers
boxplot(LungCap ~ Smoke, data = lungcapacity,
        main = "Lung Capacity by Smoking Status",
        xlab = "Smoker", ylab = "Lung Capacity (liters)")
```

`MBI` *MBI Dataset*

Description

A dataset containing gender, height and weight measurements for individuals.

Usage

```
data(MBI)
```

Arguments

A data frame object containing 10000 samples entries with the following 3 variables: "Gender", "Height", "Weight"

Name	**Type**
`Weight`	integer
`Height`	integer
`Gender`	character

Details

The dataset contains 10000 samples with three variables. We can use this dataset to explore the relationship between height and weight. And cauculate MBI (Mass Body Index) = weight/height.

Source

Data from Kaggle: https://www.kaggle.com/datasets/mustafaali96/weight-height/data

Examples

```
data(MBI)

cor(MBI$Weight, MBI$Height, use = "complete.obs")
```

`personality` *Extrovert vs. Introvert Personality Traits Dataset*

Description

Dive into the Extrovert vs. Introvert Personality Traits Dataset, a rich collection of behavioral and social data designed to explore the spectrum of human personality. This dataset captures key indicators of extroversion and introversion, making it a valuable resource for psychologists, data scientists, and researchers studying social behavior, personality prediction, or data preprocessing techniques.

Usage

```
data(personality)
```

Arguments

A data frame with 2,900 observations and the following 8 variables:

Variable	**Type**	**Description**
`AloneTime`	numeric	Daily hours spent alone (0-11 hours)
`StageFear`	character	Presence of stage fright (Yes/No)
`SocialEvents`	numeric	Frequency of social events attended (0-10 scale)
`GoOut`	numeric	Frequency of going outside (0-7 days/week)
`Drained`	character	Feeling drained after socializing (Yes/No)
`Friends`	numeric	Number of close friends (0-15 people)
`PostFreq`	numeric	Social media post frequency (0-10 scale)
`Personality`	character	Target personality classification (Extrovert/Introvert)

Source

The dataset is taken from an open source on Kaggle website https://www.kaggle.com/datasets/example/personality-traits

Examples

```
data(personality)

# Examine relationship between social events and friends count
cor(personality$SocialEvents, personality$Friends, use = "complete.obs")

# Plot distribution of social engagement
barplot(table(personality$SocialEvents),
        main = "Distribution of Social Event Attendance",
        xlab = "Attendance Frequency",
        ylab = "Number of Individuals")
```

`pollution` *Air Quality Index (AQI) for Chinese Cities (2022)*

Description

A multidimensional dataset containing weekly Air Quality Index (AQI), meteorological parameters, and socioeconomic indicators for 173 Chinese cities in 2022.

Usage

```
data(pollution)
```

Arguments

A list object containing 173 city entries with the following 10 variables:

Name	Type	Description
AQI	matrix	Air Quality Index, a matrix with 173 rows (cities) and 51 columns (weekly AQI values). Higher values indicate poorer air quality.
city	character	City names vector (length 173).
temp	numeric	Annual mean air temperature in °C.
dew	numeric	Annual mean dew point temperature in °C.
windD	numeric	Wind direction in degrees (0-360).
windS	numeric	Annual mean wind speed in m/s.
pres	numeric	Annual mean atmospheric pressure in hPa.
pop	numeric	Household resident population (unit: 10,000).
green	numeric	Green Covered Area as percentage of Completed Area (0-100).
second	numeric	Secondary Industry as Percentage to GRP (0-100).

Details

The data provides AQI data for 173 Chinese cities for the 51 weeks of 2022 and economic and meteorological related annual average data.

Source

- Air Quality Index form China National Environmental Monitoring Center(https://air.cnemc.cn:18007/)
- Meteorological Data from NOAA National Centers for Environmental Information (https://www.ncei.noaa.gov/)
- Socioeconomic data from China City Statistical Yearbook (https://www.stats.gov.cn/)

References

Guan, X., Li, Y., Liu, X., & You, J. (2025). Subgroup learning in functional regression models under the RKHS framework. *arXiv preprint arXiv:2503.01515*.

Examples

```
data(pollution)

# Explore AQI distribution for Beijing
bj_aqi <- as.numeric(pollution$AQI[pollution$city == "Beijing", ])
plot(bj_aqi,
     type = "l",
     main = "Weekly AQI in Beijing (2022)",
     xlab = "Week",
     ylab = "AQI")

# Correlation analysis
cor(pollution$temp, rowMeans(pollution$AQI, na.rm = TRUE))
```

`skcm` *Skin Cutaneous Melanoma (SKCM) data*

Description

The dataset contains clinical outcome measurements and high-dimensional gene expression profiles from 361 subjects with skin cutaneous melanoma.

Usage

```
data(skcm)
```

Arguments

`y`	A numeric vector of length 361 representing Breslow's thickness, a clinicopathologic feature of cutaneous melanoma.
`gexp`	A data frame with 361 rows and 2000 columns, where each column represents expression values of a gene. Gene names are provided as column names (e.g., SLC8A1, DPYD).
`adj_matrix`	A 2000 x 2000 adjacency matrix representing gene-gene interaction relationships in `gexp`. Entries are defined as: • 1: If two genes are directly interacting in the KEGG melanoma pathway (hsa05218) • 0: Otherwise (no interaction or gene not in the pathway) We obtain melanoma-related pathway data through KEGG portal: 1. Access https://www.kegg.jp/ and search "melanoma" 2. Identify pathway ID: `hsa05218` (Skin Cutaneous Melanoma) 3. Contains 73 genes with curated regulatory relationships Key construction steps see Details.

Details

The dataset includes 361 samples with outcomes (Breslow's thickness measurements) and expression levels of the top 2000 most variable genes. It is derived from The Cancer Genome Atlas (TCGA) for Skin Cutaneous Melanoma (SKCM), one of the most aggressive cancer types.

`adj_matrix` construction follows these key steps:

1. **Pathway Data Retrieval**: Download KGML file from KEGG via:

   ```
   download.file("https://rest.kegg.jp/get/hsa05218/kgml", "melanoma_kgml.xml")
   ```

2. **Graph Parsing**: Convert KGML to adjacency matrix:

   ```
   if (!require("BiocManager", quietly = TRUE))
   install.packages("BiocManager")
   BiocManager::install("KEGGgraph")

   library(KEGGgraph)
   melanoma_kgml <- "melanoma_kgml.xml"
   kegg_graph <- parseKGML2Graph(melanoma_kgml, genesOnly = TRUE)
   adjacent <- as(kegg_graph, "matrix")
   adj_matrix_KEGG <- adjacent + t(adjacent)
   ```

3. **Gene ID Conversion**: Map KEGG IDs to Gene Symbols:

```
BiocManager::install("KEGGREST")
BiocManager::install("org.Hs.eg.db")

library(KEGGREST)
library(org.Hs.eg.db)
kegg_genes <- keggLink("hsa", "hsa05218")
kegg_ids <- gsub("hsa:", "", kegg_genes)
gene_symbols <- mapIds(org.Hs.eg.db,
                    keys = kegg_ids,
                    column = "SYMBOL",
                    keytype = "ENTREZID")  # Get Symbol mapping
rownames(adj_matrix_KEGG) <- gene_symbols[match(rownames(adj_matrix_KEGG), kegg_genes)]
colnames(adj_matrix_KEGG) <- rownames(adj_matrix_KEGG)
```

4. **Matrix Embedding**: Create final adjacency matrix:

```
gene_skcm <- colnames(skcm$gexp)
p <- length(gene_skcm)
adj_matrix_skcm <- matrix(0, nrow = p, ncol = p, dimnames = list(gene_skcm, gene_skcm))
common_genes <- intersect(gene_skcm, gene_symbols)
adj_matrix_skcm[common_genes, common_genes] <- adj_matrix_KEGG[common_genes, common_genes]
```

Source

The Cancer Genome Atlas (TCGA) portal: https://tcga-data.nci.nih.gov

References

The Cancer Genome Atlas Consortium. (2015). Genomic classification of cutaneous melanoma. *Cell*, **161**(7), 1681-1696.

Tan, X., Zhang, X., Cui, Y., & Liu, X. (2024). Uncertainty quantification in high-dimensional linear models incorporating graphical structures with applications to gene set analysis. *Bioinformatics*, **40**(9), btae541.

Examples

```
data(skcm)
hist(skcm$y, main = "Distribution of Clinical Outcomes", xlab = "Outcome Value")
pca_result <- prcomp(skcm$gexp[, 1:100], scale = TRUE) # Run PCA on the top 100 genes
plot(pca_result$x[, 1:2], main = "PCA of Gene Expression Data")
```

---

socialmedia *Students' Social Media Addiction*

Description

This dataset contains anonymized records of students' social media behaviors and related life outcomes. Spanning multiple countries and academic levels, it captures key dimensions including usage intensity, platform preferences, and relationship dynamics.

Usage

```
data(socialmedia)
```

Arguments

A data frame with configurable observations (typically 100-1000) and the following 13 variables:

Variable & **Type** & **Description**
`id` & integer & Unique student identifier
`age` & integer & Age in years (16-25)
`gender` & character & Gender identity (Male/Female)
`academic` & character & Academic level (High School/Undergraduate/Graduate)
`country` & character & Country of residence (Bangladesh, India, USA, UK, etc.)
`usage` & numeric & Average daily social media usage in hours
`platform` & character & Most frequently used platform (Instagram, Facebook, TikTok, etc.)
`affects\_academic` & character & Self-reported academic impact (Yes/No)
`sleep` & numeric & Average nightly sleep duration in hours
`health` & integer & Self-rated mental health score (1 = poor to 10 = excellent)
`relationship` & character & Relationship status (Single/In Relationship/Complicated)
`conflicts` & integer & Number of relationship conflicts attributed to social media
`addiction` & integer & Social media addiction score (1 = low to 10 = high)

Source

The dataset is taken from an open source on Kaggle website https://www.kaggle.com/datasets/adilshamim8/social-media-addiction-vs-relationships

Examples

```
data(socialmedia)

# Visualize addiction scores by platform
boxplot(addiction ~ platform, data = socialmedia,
        main = "Addiction Scores by Social Media Platform",
        xlab = "Platform", ylab = "Addiction Score")
```

---

wage1 — *Wage Determinants Dataset (1976 U.S. CPS)*

Description

A classic dataset in econometrics for analyzing wage determinants, derived from the 1976 U.S. Current Population Survey (CPS). It contains 526 observations on 24 variables related to workers' education, experience, demographic characteristics, and hourly wages. This dataset is widely used in teaching and research on labor economics, particularly for estimating linear regression models of wage determination.

Usage

```
data(wage1)
```

Format

A data frame with 526 observations on 24 variables:

wage average hourly earnings

educ years of education

exper years of potential labor market experience

tenure years with current employer

nonwhite equals 1 if nonwhite

female equals 1 if female

married equals 1 if married

numdep number of dependents

smsa equals 1 if living in a Standard Metropolitan Statistical Area

northcen equals 1 if living in the north central U.S.

south equals 1 if living in the southern region

west equals 1 if living in the western region

construc equals 1 if working in the construction industry

ndurman equals 1 if in nondurable manufacturing

trcommpu equals 1 if in transportation, communication, or public utilities

trade equals 1 if in wholesale or retail trade

services equals 1 if in the services industry

profserv equals 1 if in professional services

profocc equals 1 if in a professional occupation

clerocc equals 1 if in a clerical occupation

servocc equals 1 if in a service occupation

lwage log of hourly wage

expersq square of `exper`

tenursq square of `tenure`

Details

This dataset focuses on how individual attributes influence labor market outcomes, with an emphasis on wage determination. It includes both continuous variables (e.g., education, experience) and binary indicators (e.g., marital status, region), making it suitable for estimating Mincer earnings functions—a foundational model in labor economics that relates log wages to human capital variables. The rich mix of socioeconomic and demographic characteristics enables analyses of wage differentials, returns to education, and experience effects.

Source

Originally from the 1976 U.S. Current Population Survey (CPS). Also available in the **Wooldridge** R package.

References

Wooldridge, J. M. (2020). *Introductory Econometrics: A Modern Approach* (7th ed.). Cengage.

Farber, H. S. (1981). The determination of the elasticity of substitution between workers in production. *Review of Economics and Statistics*, 63(4), 505–512.

Examples

```
data(wage1)

wage_model <- lm(wage ~ educ + exper + tenure + married + smsa, data = wage1)
summary(wage_model)

coef(wage_model)["educ"]

plot(wage1$educ, wage1$wage,
     xlab = "Years of Education",
     ylab = "Hourly Wage (USD)",
     main = "Wage vs. Education Level")

abline(lm(wage ~ educ, data = wage1), col = "green", lwd = 2)
```

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